

Tutorial-09

CID	Credit Score	Annual Income	Loan amount	Emp Years	Default
1	650	45	150	2	1
2	750	80	200	8	0
3	620	38	180	1	1
4	780	95	250	10	0
5	640	42	160	2	1
6	760	85	220	9	0
7	630	40	120	1	1
8	770	90	230	10	0
9	660	48	155	3	1
10	740	78	210	7	0
11	625	36	125	1	1
12	785	92	240	4	0
13	655	46	165	2	1
14	755	82	215	8	0
15	635	41	172	2	1
16	775	88	235	9	0
17	645	44	158	2	1
18	765	86	225	8	0
19	628	39	168	1	1
20	780	94	245	10	0

Step 1: Divide Dataset (k-Fold Cross Validation)

Total = 20 Samples.

Default (Target Variable) $\rightarrow$ 10 Samples

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Each fold = 4 Samples $\Rightarrow$ each fold consist of (2(1)3), (2(0)3)

Step 2:

Fold 1: Customers $\Rightarrow$ 1(1), 2(0), 3(1), 4(0)

2: Customers $\Rightarrow$ 5(1), 6(0), 7(1), 8(0)

3: Customers $\Rightarrow$ 9(1), 10(0), 11(1), 12(0)

4: Customers $\Rightarrow$ 13(1), 14(0), 15(1), 16(0)

5: Customers $\Rightarrow$ 17(1), 18(0), 19(1), 20(0)

Step 3:

Select Fold (1)

Validation Set (test) = Customers (1, 2, 3, 4)

Training Set = Customers (5 to 20)

$\hookrightarrow$ Remaining Customers

Step 4: Logistic Regression Formula.

$$P = \frac{1}{1 + e^x}$$

$$Z = w_0 + w_1 X_1 + w_2 X_2 + w_3 X_3 + \dots + w_n X_n$$

$$Z = 15 - 0.02(\text{Credit}) - 0.05(\text{Income}) + 0.03(\text{Loan}) - 0.5$$

$$1) Z = 3.25$$

$$2) Z = -2$$

$$3) Z = 5.6$$

$$4) Z = -2.85$$

Step 4: Calculate for each Validation.

Customer 1,

$$Z = 15 - 0.02(650) - 0.05(45) + 0.03(150) - 0.5(3)$$

$$Z = 3.25$$

$$P = \frac{1}{1 + e^{-Z}} = \frac{1}{1 + e^{-3.25}} = 0.962$$

$$P = 0.962 \geq 0.5 \Rightarrow \text{Default (predicted 1)}$$

Results:

Customer	Actual	Predicted	Probability
1	1	1	0.962
2	0	0	0.119
3	1	1	0.996
4	0	0	0.05

Confusion Matrix

	Pred 1	Pred 0
Actual 1	2	0
Actual 0	0	2

(or)

	Pred Yes	No
Actual Yes	TP	FN
No	FP	TN

Calculate Metrics:

$$\text{Accuracy} = \frac{\text{Correct prediction}}{\text{Total prediction}} = \frac{4}{4} = 1.00$$

$$\text{Precision} = \frac{TP}{TP+FP} = \frac{2}{2+0} = 1$$

$$\text{Recall} = \frac{TP}{TP+FN} = \frac{2}{2+0} = 1$$

$$F_1 - \text{Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Precision + Recall

Actual \ Predicted	Actual = 0	Actual = 1
Predicted = 0	2	0
Predicted = 1	0	2